

Mount Modules on a DIN Rail SOP (1x Episode: Five Modules)

One episode mounts all five modules on one rail. Work the five phases in this fixed order: **snap the five modules onto the rail in drawing order, push the row together against the start mark, fit the two end stops, screw down the five leads, press on the label strip.** The drawing, the rail, and the order of the phases never vary.

This is a single-arm task. Only the **right arm** works. The **left arm stays at home with its gripper open for the whole episode** and touches nothing. Because there is no second gripper to hold or steady anything, the fixtures do that work: the rail is screwed to a heavy **mounting plate** that is clamped to the table, so a module can be snapped on, a row pushed, a screw turned, and a label pressed without the rail moving. Nothing is held in the air while something else is done to it.

Each module is snapped on with a small gap to the right of the one before it, so the right gripper has room for its fingers. Step 2 then closes every gap with one push from the right end.

The table is set up in one of three ways. Only the **module rack**, with the five modules standing in it, moves. The mounting plate, the end stop tray, the lead rack, the driver stand, and the label tray are in the same place in all three.

- **Config L:** the module rack stands at the center-left, left of the mounting plate.
- **Config M:** the module rack stands at the back-center, behind the mounting plate.
- **Config R:** the module rack stands at the center-right, right of the mounting plate, between the driver stand and the lead rack.

Where a step depends on the setup it says so on an **IF** line. Look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the module rack stands at the center-left (**Config L**), the back-center (**Config M**), or the center-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Single arm:** the **right gripper** does all the work in all three configs, and the **left arm stays parked** even when the rack is on the left. The config changes only which way the right gripper reaches for each module. Nothing is handed over.
- **Fixed order:** the modules go on the rail in the order the drawing card shows, left to right. Every later phase also works left to right: position 1 first, position 5 last.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the mounting plate with the rail and the drawing card at the center, the module rack in the start zone for this episode's config, the end stop tray at the back-left, the label tray at the front-left, the driver stand at the back-right, the lead rack at the front-right, and the front table edge.
3. The rail is visible from above along its whole length, so the start mark, every module, both end stops, and the label strip can be seen.
4. The front faces of the modules are visible, so every lead hole and every bare end can be seen.
5. Both arms are at home with grippers open.
6. The table is clean and bare apart from the mounting plate, the module rack and modules, the end stop tray, the lead rack and leads, the driver stand and driver, and the label tray and label strip.
7. The right arm reaches the whole length of the rail, all five rack slots in all three start zones, the end stop tray, the label tray, the driver stand, and the lead rack without stretching.
8. The left arm is at home, gripper open, and stays there.

Materials checklist

1. One **mounting plate** lies flat at the center of the table, clamped down at both ends. It is heavy and does not slide or lift when a module is pressed on or a screw is turned.
2. One **rail** is screwed flat to the plate and runs left to right across its middle. Its **back lip** faces the back of the table and its **front lip** faces the front edge.
3. A **start mark** is printed on the plate just in front of the rail, near its left end.
4. The **drawing card** is taped flat on the plate behind the rail. It shows, left to right, positions 1 to 5. Under each position it shows the module tag that goes there, the lead color for it, and the name printed on the label strip above it. It also shows one end stop at each end of the row.
5. **Five modules**, each with a colored **tag** and a letter from **A** to **E** printed on its top. They are not all the same, and the drawing does not put them in A to E order.

6. The five modules stand in the **module rack**, one per slot, in A to E order left to right. Each one stands base down, with its lead hole facing the front edge, the way it will sit on the rail. The rack stands in the start zone for the config:
 - **Config L**: center-left, left of the mounting plate.
 - **Config M**: back-center, behind the mounting plate.
 - **Config R**: center-right, right of the mounting plate.
7. Each module has one **lead hole** in its front face and one **screw** on its top, just behind the front face. Every screw starts **open**: backed off far enough that a bare end slides into its hole without force.
8. Each module has a **hook** at the back of its base and a spring **catch** at the front of its base, with a small loop that faces the front edge.
9. Two **end stops** lie flat side down in the **end stop tray** at the back-left.
10. **Five leads** lie flat in the **lead rack** at the front-right, one per slot, bare end pointing left. Each lead is one color, and each color matches one module tag. The **bare end** is straight, with no kink, and about as long as the lead hole is deep.
11. One **driver** stands in the **driver stand** at the back-right, handle up, blade down. The blade fits the screw slots. The stand is heavy and does not tip when the driver is lifted out or put back.
12. One **label strip** lies printed side up in the **label tray** at the front-left, its left end arrow pointing left. It is one stiff piece carrying five names, spaced so that each name sits over one module when the five modules touch with no gaps.

Workspace layout

Everything below is a fixed area of the table, judged by eye against the table edges and the fixtures on it.

- **Center: mounting plate.** The rail across the middle, the start mark in front of its left end, the drawing card behind it.
- **Start zone: module rack.** Center-left (Config L), back-center (Config M), or center-right (Config R). Five modules standing in A to E order.
- **Back-left: end stop tray.** Two end stops.
- **Front-left: label tray.** The label strip.
- **Back-right: driver stand.** The driver, handle up.
- **Front-right: lead rack.** Five leads, bare ends pointing left.

Arm assignment

- **Right gripper: all of the work.** Snaps each module onto the rail, pushes the row together, fits both end stops, puts each lead into its hole, turns every screw with the driver, tug-tests every lead, and presses on the label strip, in all three configs.

- **Left arm: parked.** Stays at home with its gripper open from the start of recording to the end, in all three configs. It holds nothing, steadies nothing, and reaches for nothing.

Vocabulary

- **Rail:** the metal strip screwed to the mounting plate. Every module and end stop snaps onto it.
- **Back lip / front lip:** the two raised edges of the rail. The back lip is toward the back of the table, the front lip toward the front edge.
- **Mounting plate:** the heavy plate the rail is screwed to. It is clamped to the table and never moves.
- **Start zone:** where the module rack stands at the start of the episode: center-left (**Config L**), back-center (**Config M**), or center-right (**Config R**).
- **Drawing card:** the card taped behind the rail. It says which module goes in which position, which lead goes in each module, and which name sits over it.
- **Position:** one place in the row, numbered 1 at the left to 5 at the right on the drawing card.
- **Tag:** the colored mark and letter (A to E) on top of each module.
- **Start mark:** the line printed in front of the left end of the rail. The left side of the position 1 module lines up with it.
- **Hook:** the lip at the back of a module's base. It goes over the back lip of the rail first.
- **Catch:** the spring clip at the front of a module's base. It clicks over the front lip of the rail.
- **Snapped on:** the hook is over the back lip, the catch has clicked over the front lip, and the base sits flat on the rail. The module does not rock when touched from the front.
- **Gripper gap:** the small gap left to the right of a module when it is snapped on, just wide enough for the gripper finger. It is closed in Step 2.
- **Touching:** two modules, or a module and an end stop, sit side against side with no light showing between them from above.
- **In line:** seen from above, the front faces of all five modules make one straight line along the rail.
- **End stop:** the small block that snaps onto the rail at each end of the row and stops the modules sliding. Its **flat side** faces the modules.
- **Lead:** one colored wire with a bare end.
- **Bare end:** the short piece of bare metal at one end of a lead.
- **Matching lead:** the lead whose color is the color shown for that position on the drawing card.
- **Lead hole:** the opening in the front face of a module that the bare end goes into.
- **Fully home:** the bare end is all the way in the lead hole and the colored cover of the lead touches the front face. No bare metal shows outside the hole.

- **Screw slot:** the straight groove across the top of a screw that the driver blade sits in.
- **Driver:** the screwdriver used to turn the screws. It lives in the driver stand when it is not in the right gripper.
- **Blade seated:** the driver blade sits down in the screw slot, in line with it, and does not slip out when turned.
- **Quarter turn:** one turn of the driver a quarter of the way round, clockwise.
- **Closed screw:** the screw has stopped turning under a quarter turn and holds the lead tight.
- **Tug test:** one short, gentle pull on a lead toward the front edge, to show the screw holds it.
- **Label strip:** the stiff printed strip that carries the five names. Its **left end arrow** points to the left end of the row.
- **Label groove:** the groove across the top of each module, behind the screw, that the label strip clicks into.
- **Pressed home:** the label strip sits down flat in the label groove of all five modules, with no end or middle standing up.
- **Stable placement:** the item stays still for 2 seconds after the right gripper lets go and does not rock, roll, tip, or slide.
- **Parked arm:** the left arm, at home with its gripper open. It does not move during the episode.

Steps

Only Step 1.1 depends on where the module rack stands. Every other line is the same in all three configs.

Step 1: Snap five modules onto the rail

Goal: all five modules are snapped on in drawing order, left to right, each with a gripper gap to its right.

Work the positions in this fixed order: **position 1, then 2, 3, 4, 5**. Run both parts of this step on one module before starting the next. The **right gripper** does every action.

1.1 Take the next module

- Read the drawing card: find the tag shown for this position.
- With the **right gripper**, grasp that module across its left and right sides, near the top.
- With the **right gripper**, lift it straight up out of its rack slot, base down, lead hole toward the front edge.
- **IF the rack is at the center-left (Config L):** carry the module straight right to the rail.
- **IF the rack is at the back-center (Config M):** carry the module forward, low over the drawing card, to the rail.
- **IF the rack is at the center-right (Config R):** carry the module straight left to the rail.

- Then, in all three: with the **right gripper**, bring the module over its place on the rail. For position 1 that is the start mark. For every other position it is just right of the module before it, with a gripper gap between them.

Check: the module in the **right gripper** carries the tag the drawing card shows for this position, and its lead hole faces the front edge. If the tag is wrong, put it back in its own rack slot with the **right gripper** and take the right one.

1.2 Snap it on

- With the **right gripper**, tip the module slightly back and lower it until the hook sits over the back lip of the rail.
- With the **right gripper**, rock the front of the module down toward the front lip until the catch clicks.
- With the **right gripper**, let go and lift the gripper clear.
- For position 1, the left side of the module lines up with the start mark.

Check: the module is snapped on: it sits flat, does not rock, and has a gripper gap to its right. If the catch did not click, press the front of the module down once more with the closed **right gripper**. If the module is on the rail in the wrong position or the wrong way round, set the driver blade in the catch loop with the **right gripper**, pull the blade toward the front edge until the front lifts, tip the module off the back lip, put the driver back in its stand, and snap the module on again the right way.

Repeat Step 1 for positions 2, 3, 4, and 5. Only when all five modules are on the rail does Step 2 start.

Expected state: five modules on the rail in drawing order, a gripper gap between each pair, the position 1 module on the start mark. The module rack is empty.

Step 2: Push the row together

Goal: the five modules touch each other with no gaps, sit in line, and the position 1 module is still on the start mark.

- With the **right gripper** closed, set it against the right side of the position 5 module, low on the side.
- With the **right gripper**, push slowly to the left along the rail. Each module slides until it touches the next one.
- Stop pushing as soon as all five modules touch. Do not push further once position 1 would start to move.
- With the **right gripper**, lift clear.

Check: all five modules touch, they are in line, and the left side of position 1 still lines up with the start mark. If a gap is left, push once more from the right side of position 5 with the closed

right gripper. If position 1 has moved off the start mark, push it back with the closed **right gripper** from its left side, then close the row again from the right.

Expected state: one solid row of five modules on the rail, left side on the start mark.

Step 3: Fit the two end stops

Goal: one end stop sits touching each end of the row, flat side toward the modules, snapped on.

Fit the **left end stop first, then the right one.**

- With the **right gripper**, grasp one end stop from the end stop tray across its sides.
- With the **right gripper**, turn it so its flat side faces the row.
- With the **right gripper**, snap it onto the rail the same way as a module (hook over the back lip, rock the front down until it clicks), with a gripper gap between it and the row.
- With the **right gripper** closed, push the end stop along the rail until its flat side just touches the row. Stop as soon as it touches. The row must not move.
- Take the second end stop with the **right gripper** and fit it at the right end of the row the same way.

Check: each end stop is snapped on, touches the row, and faces it with its flat side. Position 1 is still on the start mark and no gap has opened in the row. If an end stop is not touching, push it once more with the closed **right gripper**.

Expected state: the row is locked between two end stops. The end stop tray is empty.

Step 4: Put the five leads in

Goal: each module has its matching lead fully home in its lead hole. Every screw is still open.

Work the positions in this fixed order: **position 1, then 2, 3, 4, 5.**

- Read the drawing card: find the lead color shown for this position.
- With the **right gripper**, grasp that lead on its colored cover, just behind the bare end. Never grasp the bare end.
- With the **right gripper**, lift it out of the lead rack and carry it to the front of the module, bare end pointing back.
- With the **right gripper**, line the bare end up with the lead hole and push it straight in until it is fully home.
- With the **right gripper**, let go and lift clear. The rest of the lead lies on the plate toward the front edge.

Check: each lead is the matching lead for its position and is fully home, with no bare metal showing. If bare metal shows, push the lead in again with the **right gripper**. If a lead is in the wrong module, pull it straight out with the **right gripper** and put it in the right one.

Expected state: five leads fully home, one per module, all screws still open. The lead rack is empty.

Step 5: Screw the five leads down

Goal: all five screws are closed on their leads, turned with the driver, left to right, and every lead passes a tug test.

The **right gripper** turns screws with the **driver** only, never with its own tip.

5.1 Take the driver

- With the **right gripper**, pinch the driver by its handle and lift it straight up out of the driver stand.
- With the **right gripper**, hold the driver blade down, upright.

5.2 Close each screw

Close **position 1 first, then 2, 3, 4, 5**. Do not skip a screw and do not come back out of order.

- Look first: the lead for this module is still fully home. If it has slid out, put the driver back in its stand with the **right gripper**, push the lead home, and take the driver again.
- With the **right gripper**, set the blade down in the screw slot, in line with the slot.
- With the **right gripper**, turn the driver one quarter turn clockwise.
- With the **right gripper**, lift the blade clear, set it back down in the slot, and turn another quarter turn.
- Repeat quarter turns with the **right gripper** until the screw stops turning. Never force a screw once it has stopped.
- If the blade slips out of the slot, lift it clear with the **right gripper**, set it back in the slot, and turn again.

5.3 Put the driver back

- With the **right gripper**, carry the driver upright to the driver stand, lower it in blade first, and let go when it stands on its own.

5.4 Tug-test each lead

Test **position 1 first, then 2, 3, 4, 5**, with the driver already in its stand.

- With the **right gripper**, grasp the lead on its colored cover, a little in front of the module.
- With the **right gripper**, give one short, gentle pull toward the front edge.
- With the **right gripper**, let go.

Check: every screw is closed and every lead stays fully home after its tug test. If a lead moves or comes out, push it home with the **right gripper**, take the driver again, close that screw, put the driver back, and tug-test that lead once more.

Expected state: five leads held tight, the driver standing in its stand, the row and end stops unmoved.

Step 6: Press on the label strip

Goal: the label strip is pressed home in the label grooves, each name over the module the drawing card shows it with.

- With the **right gripper**, grasp the label strip across its middle, printed side up.
- With the **right gripper**, lift it out of the label tray, keeping it flat, left end arrow pointing left.
- With the **right gripper**, carry it over the row and lower it onto the label grooves, left end over position 1.
- With the **right gripper**, let go.
- With the **right gripper** closed, press down on the strip over position 1 until it clicks, then press over positions 2, 3, 4, and 5 in turn.

Check: the strip is pressed home along its whole length, the names read left to right from the front, and each name sits over the module shown with it on the drawing card. If an end or the middle stands up, press it down again with the closed **right gripper**. If the strip is the wrong way round, lift it off with the **right gripper** and lay it on again with the arrow pointing left.

Expected state: the label strip is pressed home. The label tray is empty.

Step 7: End the episode

Goal: recording ends with the five modules mounted, locked, wired, and labelled.

- Look once along the rail: five modules touching in drawing order from the start mark, an end stop at each end, five leads screwed down, the label strip pressed home, and the driver in its stand.
- Return the **right arm** home with its gripper open. The left arm is already home. Homing is the last thing the arm does.
- Stop recording.

Check: both arms are at home, both grippers are open, and neither holds anything.

Expected state: the row is finished and still, the mounting plate has not moved, and the recording has stopped.

After the episode: reset the workspace

This reset is not recorded.

1. Pull the label strip off and lay it printed side up in the label tray, left end arrow pointing left.
2. Open each screw with the driver until its lead slides out. Lay the five leads in the lead rack, bare ends pointing left. Leave every screw open.

3. Take both end stops off the rail and lay them in the end stop tray.
4. Take the five modules off the rail and stand them in the module rack in A to E order, base down, lead hole facing the front edge.
5. Stand the module rack in the start zone for the next episode's config: center-left (Config L), back-center (Config M), or center-right (Config R).
6. Stand the driver in the driver stand, handle up.
7. Check every bare end. Straighten or replace any lead with a bent or frayed bare end.
8. Check that the mounting plate is still clamped and the drawing card and start mark are readable. Re-tape the card if it is lifting.
9. Inspect the modules, catches, end stops, screws, and label strip for damage. Replace damaged items.
10. Run both Setup checklists again.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain several violations; tag each one separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Violation: Config misaligned

- **Visible cue:** the episode does not match the config on the table: the module rack is not in the start zone for the config, the right gripper carries a module on the path for a different config, or the wrong IF line is followed.
- **SOP rule broken:** Step 1.1, the module rack stands at the center-left, back-center, or center-right for the episode's config, and the IF line followed is the one for the config on the table.

- **Coaching note:** look where the rack is before the first reach, then follow that config's IF line in Step 1.1.

Violation: Module in the wrong position

- **Visible cue:** a module is snapped on in a position whose tag on the drawing card does not match it, for example module A in position 1 when the card shows C there; or the modules are taken in A to E rack order instead of drawing order.
- **SOP rule broken:** Step 1.1, each position takes the module whose tag the drawing card shows for it.
- **Coaching note:** read the card for the position first, then look for that tag in the rack.

Violation: Module on the wrong way round

- **Visible cue:** a module is snapped on with its lead hole facing the back of the table, or is turned in the right gripper during the carry.
- **SOP rule broken:** Step 1.1 (lift each module base down, lead hole toward the front edge, and keep it that way to the rail).
- **Coaching note:** the rack holds each module the way it sits on the rail. Lift it straight up and do not turn it.

Violation: Module not snapped on, or snapped on wrong

- **Visible cue:** a module is released with its catch not clicked, sitting tilted, or rocking; the module is pressed straight down or forced without the hook going over the back lip first; or the right gripper pulls a module off the rail by hand instead of with the driver in the catch loop.
- **SOP rule broken:** Step 1.2 (hook over the back lip first, rock the front down until the catch clicks, and take a module off only with the driver blade in the catch loop).
- **Coaching note:** back lip first, then rock down. Listen and look for the click before letting go.

Violation: No gripper gap left

- **Visible cue:** a module is snapped on touching the one before it, so the right gripper finger is pinched or bumps the module already on the rail; or position 1 is snapped on away from the start mark.
- **SOP rule broken:** Step 1.1 and 1.2 (snap each module just right of the one before it with a gripper gap, and position 1 on the start mark).
- **Coaching note:** leave room for the finger. The gaps are closed in Step 2, not during the snap.

Violation: Work order wrong

- **Visible cue:** positions are snapped on, wired, screwed, or tug-tested out of left to right order; or a phase starts before the one before it is finished, such as an end stop fitted before

the row is pushed together, a lead put in before both end stops are on, a screw turned before all five leads are in, or the label strip pressed on before every lead passes its tug test.

- **SOP rule broken:** Steps 1 to 6, work the phases in the fixed order (snap, push together, end stops, leads, screws, label strip), and within each phase work position 1 to position 5.
- **Coaching note:** finish each phase and its check before starting the next one, and always start at the left.

Violation: Row not pushed together or not in line

- **Visible cue:** a gap still shows between two modules when Step 3 starts, the fronts are not in one line, or position 1 has left the start mark.
- **SOP rule broken:** Step 2 (push until all five modules touch, in line, with position 1 on the start mark).
- **Coaching note:** look along the row from above before reaching for an end stop.

Violation: Row pushed wrong

- **Visible cue:** the right gripper pushes from the middle of the row instead of from the right side of position 5, pushes high on a module so it tips, lifts a module off the rail to move it, or keeps pushing so the whole row slides past the start mark.
- **SOP rule broken:** Step 2 (push with the closed gripper low on the right side of position 5 and stop as soon as all five touch).
- **Coaching note:** one slow push from the right end, low down, and stop the moment the last gap closes.

Violation: End stop fitted wrong

- **Visible cue:** an end stop is missing, is left with a gap to the row, faces the row with its shaped side, is not snapped on, is fitted right before left, or is pushed so hard the row moves.
- **SOP rule broken:** Step 3 (fit the left end stop then the right, flat side toward the row, snapped on, and pushed until it just touches).
- **Coaching note:** flat side in, snap it on with a gap, then a gentle push until it just touches.

Violation: Wrong lead in a module

- **Visible cue:** a lead is pushed into a module whose position on the drawing card shows a different lead color.
- **SOP rule broken:** Step 4, each module takes the lead whose color the drawing card shows for its position.
- **Coaching note:** read the lead color on the card before reaching for the lead rack.

Violation: Lead not fully home

- **Visible cue:** bare metal shows between the colored cover and the front face of a module when its screw is turned, or the colored cover is pushed into the hole.
- **SOP rule broken:** Steps 4 and 5.2 (push the bare end straight in until the colored cover touches the front face, and look that it is still home before turning the screw).
- **Coaching note:** push until the cover meets the face, and look again before the blade goes down.

Violation: Lead handled wrong

- **Visible cue:** the right gripper grasps a lead by its bare end, bends or kinks the bare end, drags a lead across the plate instead of lifting it, or pokes the bare end into the hole at an angle.
- **SOP rule broken:** Step 4 (grasp the lead on its colored cover just behind the bare end, lift it, and push the bare end straight in).
- **Coaching note:** hold the cover, not the metal, and line the bare end up with the hole before pushing.

Violation: Screw turned wrong

- **Visible cue:** a screw is turned with the gripper tip instead of the driver, the driver is turned more than a quarter turn in one grip, turned the wrong way, turned with the blade out of the slot, or forced on after the screw stopped.
- **SOP rule broken:** Step 5.2 (seat the blade in the slot, turn one quarter turn clockwise per grip, and stop when the screw stops turning).
- **Coaching note:** seat the blade, one quarter turn, lift, and seat again. When it stops, it is done.

Violation: Screw left open

- **Visible cue:** a screw is skipped, or the driver moves on while the screw still turns under a quarter turn.
- **SOP rule broken:** Step 5.2 (turn quarter turns on every screw until it stops turning).
- **Coaching note:** try one more quarter turn before moving on. If it still turns, it is not closed.

Violation: Driver misused or not put back

- **Visible cue:** the driver is laid on the plate or the table instead of stood in its stand; a tug test is done with the driver still in the right gripper; the driver is carried blade up; or the driver is left out of its stand at the end of the episode.
- **SOP rule broken:** Steps 5.1, 5.3 and 5.4 (lift the driver from its stand, carry it upright blade down, and stand it back in its stand before the tug tests).
- **Coaching note:** the driver is either turning a screw or standing in its stand, never lying down.

Violation: Tug test skipped or done wrong

- **Visible cue:** a lead is not tug-tested, is pulled hard enough to bend it or move the module, or is pulled by its bare end; or a lead that moved is not screwed down again and tested once more.
- **SOP rule broken:** Step 5.4 (one short, gentle pull on each lead toward the front edge, on its colored cover, and fix any lead that moves).
- **Coaching note:** a tug test is a check, not a pull out. One short, gentle pull on the cover.

Violation: Label strip applied wrong

- **Visible cue:** the strip is laid on the wrong way round or upside down, a name sits over the wrong module, the strip is dropped onto the row instead of lowered, pressed from the right end first, or left with an end or the middle standing up.
- **SOP rule broken:** Step 6 (carry the strip flat, arrow left, lower it with the left end over position 1, and press it home from position 1 to position 5).
- **Coaching note:** find the arrow before lifting. Lower it flat, then press left to right until every module clicks.

Violation: Mounting plate or rail moved

- **Visible cue:** the mounting plate shifts, lifts, or turns on the table, or the right gripper grips, pulls, or leans on the plate or the rail.
- **SOP rule broken:** Steps 1 to 6, the plate stays clamped where it is and the rail is never gripped or pushed.
- **Coaching note:** press on the module, the end stop, or the strip, never on the plate. If the plate moves, stop and report the clamp.

Violation: Parked arm moved

- **Visible cue:** the left arm leaves home, its gripper closes, or it reaches toward, holds, or steadies anything at any point in the episode.
- **SOP rule broken:** Steps 1 to 7, this is a single-arm task. The left arm stays at home with its gripper open from the start of recording to the end.
- **Coaching note:** the fixtures do the holding. If a step feels like it needs a second gripper, stop and report the fixture, do not use the left arm.

Violation: Item dropped or knocked over

- **Visible cue:** a module, end stop, lead, the driver, or the label strip falls from the right gripper onto the plate, the table, or the floor; the module rack or the driver stand is knocked over or out of place.
- **SOP rule broken:** Steps 1 to 6 (lift each item clear, carry it on the path its step names, and let go only when it is a stable placement).

- **Coaching note:** slow down near the rack and the stand, and let go only once the item is on the rail or in its place.

Violation: Finished work disturbed later

- **Visible cue:** a module already snapped on is knocked off or tipped, a closed row opens a gap, an end stop is pushed away, or a lead already home is pulled out or bent while a later step is worked.
- **SOP rule broken:** Steps 2 to 7, once an item is in its place on the rail it stays there, untouched, except for the fix its own step names.
- **Coaching note:** come down to each new place from straight above and keep the arm clear of the leads lying toward the front edge.

Violation: Check skipped

- **Visible cue:** the right arm moves straight on with no look at the step's end state: no look at the drawing card before a pick, no look for the click after a snap, no look along the row after the push, no look at the bare ends before screwing, or no look along the label strip after pressing.
- **SOP rule broken:** Steps 1 to 6, each step's check is done before the next step starts.
- **Coaching note:** every step ends with a look, and the fix happens in that step, not later.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a module missing or in the wrong position, a gap in the row, an end stop missing, a lead out or a screw open, the label strip not pressed home, or the driver out of its stand; or the right arm is away from home, a gripper is closed, or an arm makes a correction after homing.
- **SOP rule broken:** Step 7 (look once along the rail, return the right arm home with its gripper open, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arm does.

Non-violation failures

Failures that are not caused by how the task was run do not go in the violation set. Log them as system issues, discard the episode, and do not use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, collision caused by controller error, or motor error.
- **Defective item:** a module whose catch will not click on a sound rail, a stripped screw slot, a screw that will not open or close, a lead with a broken bare end, a label strip that will not

click into the grooves, or a mounting plate clamp that lets go during the episode. Replace before the next episode.

Annotation subtasks (from SOP)

1. Move one module from the rack to its position on the rail
2. Snap one module onto the rail
3. Push the row together
4. Fit one end stop
5. Put one lead into its module
6. Take the driver from the stand
7. Close one screw with the driver
8. Put the driver back in the stand
9. Tug-test one lead
10. Press the label strip onto the row
11. Return the right arm home and end the episode